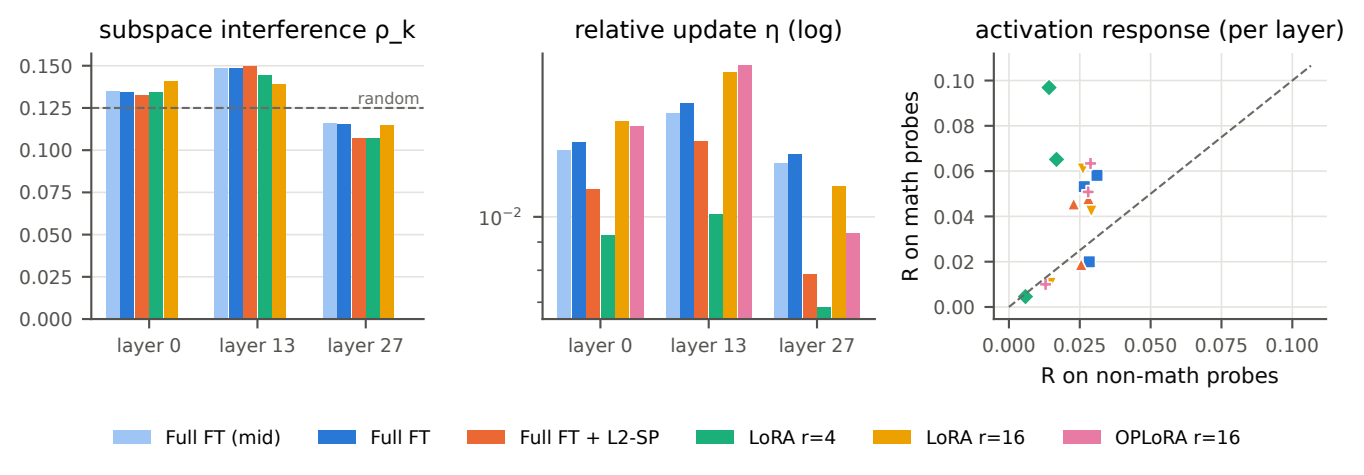
subspace interference ρ_k
0.150
0.125
0.100
0.075
0.050
0.025
0.000
random
layer 0
layer 13
layer 27
relative update η (log)
10^{-2}
layer 0
layer 13
layer 27
activation response (per layer)
R on math probes
0.10
0.08
0.06
0.04
0.02
0.00
0.000
0.025
0.050
0.075
0.100
R on non-math probes
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA r=4
LoRA r=16
OPLoRA r=16